Budget Guard: Calibrated Resource Control for Long-Horizon LLM Agents

Anonymous Author(s)
Affiliation
Address
email

Abstract

Long-horizon LLM agents cannot say how much more they need. Asked mid-task for an interval on its remaining tokens, Claude Opus 4.7 contains the truth 51%, 23%, and 35% of the time on Sokoban, Search-R1, and SWE-bench, against a 90% target. Budget Guard supplies that number from outside the agent. It reads the per-turn usage counts every API already reports, fits a small model to them, and conformally calibrates the model so that its interval contains the truth as often as promised; nothing in the agent is trained or changed. Budget awareness is two questions, how much more and whether to stop, and we build a separate calibrated instrument for each. The intervals, on Claude Opus 4.7, reach 0.90–0.96 coverage offline and 0.87–0.92 online with no unbounded intervals, and are sharper than a feature-free baseline on two of three environments. Across 19 model–environment runs, one split each, they hold at least 0.85 in 18 and at least 0.90 in 11. In distribution that much is guaranteed; the experiments map what happens outside it. Moved from Sokoban to Search-R1 and SWE-bench without recalibration, coverage falls to 0.51 and 0.10. Recalibrating from finished runs recovers 0.71 and 0.80, still below target, and after a model swap no bounded method reaches it. With three budgets at once, per-resource calibration covers the joint event 80% of the time; a Bonferroni correction restores 0.95 in distribution and falls to 0.71 under a model swap. The gate is judged on money. In Warehouse replay it avoids 86.3% of the spend on runs that could not finish, stops no run that would have finished, and cuts cost per success from \$1.61M to \$0.87M in simulated dollars; no hard cap comes close. On token tasks it saves only 7.1% and 3.1%, with 5.8% false aborts on Sokoban, and feeding the intervals into the stop rule only makes it more conservative. The intervals measure; the gate decides. All stopping results are counterfactual replays of logged runs.

1 Introduction

LLM agents now take on tasks that run for many steps, many tool calls, and many interactions with an environment [1], and the resources they consume grow with them [2]. A single attempt can burn thousands of tokens, real time, and real money before it ends [3], much of it on reasoning that changes nothing [4–6]; when the attempt fails, nearly all of it was wasted [3]. Keeping that spending under control without giving up task performance is one of the practical obstacles to running agents at scale [7–9].

The obvious fix is to ask the agent, and it does not know. BAGEN [13] asked agents mid-task for an interval on how many more tokens they would need. For Claude Opus 4.7 those intervals contain the true number 51%, 23%, and 35% of the time on Sokoban, Search-R1, and SWE-bench, against a 90% target. Figure 1 shows what that looks like on one run: the agent's interval looks precise and sits above the truth from the second turn on, reserving 400 to 900 tokens for a run that needed 267.

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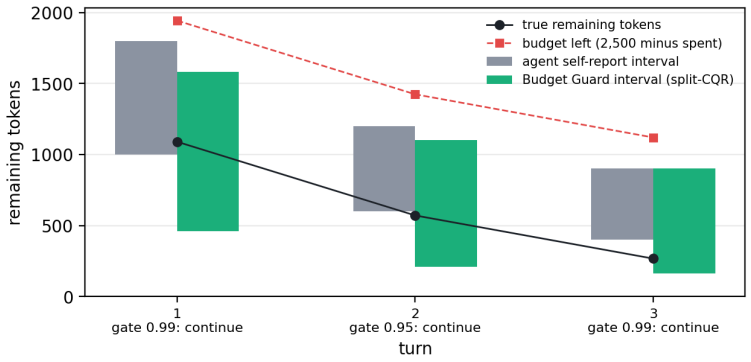


Figure 1: What Budget Guard does on one real run: the first held-out Sokoban rollout in our seed-0 test stream (Claude Opus 4.7, 2,500-token budget), chosen by that rule rather than by hand. Grey bars are the agent's own interval on the tokens it still needs, green bars the guard's, black dots the truth, and the dashed line the budget still available. The agent's interval misses from turn 2; the guard's contains the truth at every turn and stays under the remaining budget, and the guard's separate stop-or-continue gate (scores under each turn) says continue throughout. The run succeeded inside budget. Table 6 in the appendix gives the numbers.

Budget Guard supplies the missing number from outside the agent. It is a wrapper: it reads the per-turn usage that provider APIs already report, fits a small gradient-boosted model to it, and wraps that model in a conformal calibration layer, so that its interval on the remaining requirement contains the truth as often as promised. Nothing in the agent is trained or changed, and because the guard reads only usage counts it works for closed models whose internals are unavailable. In Figure 1 the green intervals are wider than the agent's and right every time; spend so far plus their upper bound stays under the budget, so a deployer could have certified from the first turn that the run fit. That is the first half of budget awareness: how much more.

The second half is whether to stop, and it is a different question with a different instrument. A feasibility gate on the same features scores each turn and aborts a run when the score drops below a threshold calibrated so that at most a chosen fraction δ of runs that would have finished are stopped. We keep the two apart throughout, with separate calibration targets, separate metrics, and separate claims, because the results say we must: the intervals are a reliable measurement, and using them as a stopping signal only makes the gate more timid.

Two facts about the label shape everything. The resources a run still needs are known only when it ends, so a method that recalibrates online may learn a run's label only once that run has finished; revealing it turn by turn would hand the guard information it cannot have in deployment, and inflates coverage (Appendix C). And the label does not exist at all for a run that can no longer finish, which is exactly the run a guard exists to catch. Half of the Warehouse turns we evaluate belong to runs in that state.

Our findings, in order. (1) In distribution the intervals do what the construction promises, with no unbounded intervals and sharper bounds than a feature-free baseline on two of three environments; the agent's own estimate adds nothing as a feature (§5.1). (2) A frozen guard collapses under an environment change and rollout-safe recalibration recovers most of the loss with bounded intervals; the methods that recover more do so by emitting unbounded intervals, and nothing bounded reaches the target after a model swap (§5.2). (3) When several budgets bind at once, per-resource calibration under-covers the joint event, and a Bonferroni correction restores it in distribution but not under swap (§5.3). (4) The gate avoids most doomed-run spend in Warehouse replay with no false aborts, which no hard cap achieves; on token tasks it saves far less, and adding the intervals to the rule makes it worse (§5.4). (5) Runs that cannot finish carry information that dropping them throws away (§5.5). Table 4 says which of these are guarantees and which are observations, and every number regenerates from one command over a pinned dataset.

2 Related work

BAGEN [13] defines budget awareness as an agent's ability to estimate mid-execution what it still needs, and asks the agent directly for an interval on remaining tokens and a judgment of whether finishing is still feasible, on Sokoban, Search-R1, SWE-bench, and Warehouse. Its central finding is a gap between capability and self-knowledge. Stopping early on the agent's own signals saves roughly 28–64% of tokens on failed trajectories at a cost of roughly 1.6–4.2 percentage points of task success, and interval coverage reaches only 47% even after supervised fine-tuning and reinforcement learning [13]. The agent carries some signal about its future; its raw estimate is not reliable enough to act on. Conformal prediction [14, 15] is the standard tool for turning an unreliable predictor into an interval with a stated coverage rate, and that is what we use it for.

Doomed From the Start [16] reads a model's internal activations to decide that a trajectory is doomed and stops it early. Like us, and unlike BAGEN, it uses a distribution-free calibrated gate with no fine-tuning, meeting recall targets of 90–97% on TextCraft and saving 47.1% (Qwen-2.5-7B) and 37.2% (Llama-3.2-3B) of inference compute at the 90% target. Because it needs internal states it applies only to open-weight models; Budget Guard needs only usage counts. MiCP [17] applies conformal prediction to a different object, the agent's answer, allocating an error budget across the turns of a multi-turn reasoning process so the agent can stop once it has enough evidence. We calibrate coverage over the remaining resource instead, sit outside the reasoning loop, and keep the abort gate separate from the intervals. Section 3 specifies the construction; Appendix A defines the terms used to read the results.

3 Method

The interval. Take any predictor, measure how badly it missed on data it never trained on, and widen its output by a high quantile of those misses. We fit two quantile regressors, at $\alpha/2$ and $1-\alpha/2$, giving a band $[\hat{f}_{\mathrm{lo}}(x), \hat{f}_{\mathrm{hi}}(x)]$; on each of n calibration points we record how far the truth fell outside it, $s = \max(\hat{f}_{\mathrm{lo}} - y,\ y - \hat{f}_{\mathrm{hi}})$, negative when the point sat inside; and with $\hat{q}$ the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of those scores, the interval $[\hat{f}_{\mathrm{lo}}(x) - \hat{q},\ \hat{f}_{\mathrm{hi}}(x) + \hat{q}]$ contains y with probability at least $1-\alpha$ [10]. We use α=0.10 and clip the lower endpoint at zero. Two things about this guarantee matter for everything that follows. It is a hit rate over runs, not a property of any one interval: across exchangeable runs the interval contains the truth at least 90% of the time, and it says nothing about which 10% it misses. And it asks nothing of the data except that the calibration runs and the deployment runs be exchangeable. Everything difficult in this paper comes from the fact that they often are not. Table 4 records, for each component, exactly what we claim and under what assumption.

What the guard sees. At turn k the guard sees only the past, the prefix of the run through turn k (Figure 2): how far along the run is (the turn index); how much it has consumed (spend so far, and spend per turn, the burn rate); how it has been consuming recently (the last turn, the mean of the last three, and the mean, standard deviation, minimum, maximum, and trend of per-turn spend); and the agent's own estimate when it gives one. Nothing that depends on the future (final totals, success, feasibility) enters the feature matrix; we assert this in code rather than trusting ourselves to have left it out. No weights, no logits, no environment internals: only the usage accounting that provider APIs already return, which is what makes the guard a wrapper rather than a modification, and a verifier rather than a believer. It never has to trust the agent's own estimate: that estimate is at most one more feature, and the guarantee holds whether or not it is any good. The predictor behind the band is gradient-boosted quantile regression [27], so the width varies with the input. We split calibration and test by rollout (one complete run), never by prefix: two prefixes of one run share its final total, and splitting them apart would leak each test answer into calibration.

Recalibrating as the stream moves. Frozen split-CQR is valid while calibration and deployment are exchangeable. Swap the underlying model, or point the guard at a new task type, and the resemblance is gone; so is the guarantee. Our response is to let the deployment stream recalibrate the guard from the runs that have already finished, in three ways. ACI [11] nudges its working confidence level up when it has been missing too often and down when it has been too safe. DtACI [12] runs several such nudges at different speeds and leans on whichever is currently working. Windowed-CQR keeps

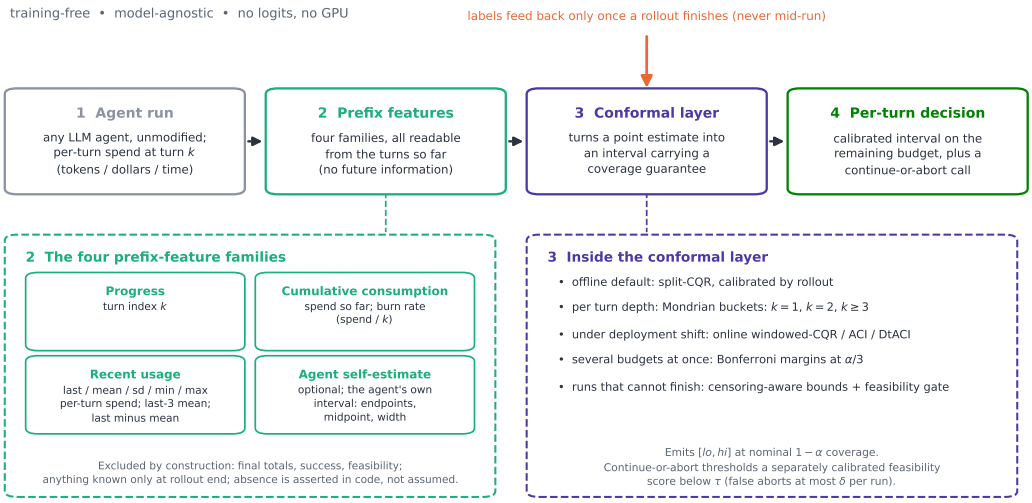


Figure 2: Budget Guard wraps an unmodified agent in an external calibration layer. Stages 1–4 share prefix features; stage 3 returns a calibrated remaining-budget interval (§§5.1–5.3), and a separately calibrated feasibility score drives continue-or-abort (§5.4). Warehouse abort economics use the feasibility gate alone and do not read the intervals. Labels become available only when a rollout finishes, so online recalibration never uses a still-running run.

a sliding window of the 100 most recent scores and reads $\hat{q}$ straight off them; the idea is borrowed from SPCI [18], though we take the plain empirical quantile rather than fitting their model of the residual series, and the honest statement of what it provides is a coverage gap that grows with how far the window has drifted from the stream [19]. All three see a label only when its rollout ends, and hold their state fixed within a run. Two reporting rules follow. An interval with no upper bound covers everything and tells you nothing, so the fraction of unbounded intervals sits beside every coverage number. And because one margin is set by the many easy late-turn prefixes, a good average can hide a hole at turn one, where an abort decision is worth the most; we therefore also calibrate a separate $\hat{q}$ per turn-depth bucket, $k{=}1$, $k{=}2$, $k{\geq}3$ (Mondrian [20]).

Several budgets, and runs that cannot finish. A Warehouse run can fail three ways, out of time, capacity, or money, and it survives only if all three estimates hold. Covering each 90% of the time does not cover all three 90% of the time, because the misses accumulate; calibrating each at miscoverage $\alpha/3$ (Bonferroni) restores the joint guarantee by the union bound at the price of wider margins, and is our primary scheme. A tighter alternative shares one threshold across resources by ranking their errors jointly [21], but that rank is chosen on the same errors it then thresholds, so it carries no guarantee; we report it as an observation, and let both move under a model swap from a rolling window of finished runs. When a run is doomed we never learn y, only that it exceeded what was left, and dropping such runs removes exactly the hard cases. We calibrate one-sided upper bounds five ways (censored scores as $+\infty$, always valid and vacuous once censoring exceeds α; the known lower bound substituted; censored runs dropped, the naive comparison; reweighting by censoring probability, IPCW [22, 23]; and charging a censored run only against the resource that stopped it), and a classifier on the same features, trained to predict whether a run can still finish, gives a feasibility score on which the guard can decline to issue an interval at all. That changes what we can promise: coverage among the prefixes the gate lets through, not across all prefixes. It is weaker than a marginal guarantee, but it is what the construction supports, so it is what we claim.

The gate. The same feasibility score, now read per run rather than per prefix, decides when to quit: abort at the first turn where it drops below a threshold τ. Since a run is aborted the moment its lowest score crosses τ, we set τ from the lowest scores of the feasible calibration runs, taking their δ-quantile with the finite-sample correction above, which bounds the probability of aborting a run that would have succeeded by δ, a per-run guarantee in the sense of conformal risk control [26]. Aborting a doomed run recovers what it had left to spend; aborting a good one throws away a success. Both land in the same currency, so the gate is judged on Cost-of-Pass [25], total spend per successful

Table 1: The four evaluation environments; only Warehouse binds several resources at once, and only Warehouse censors labels. Token budgets are the BAGEN context budgets in tokens; Warehouse budgets are set per task. In Warehouse, half of the prefixes (640 of 1,280 per model) belong to runs that can no longer finish and drive the censored-calibration evaluation.

Environment	Task	Budgeted resource(s)	Budget	# Models
Sokoban	box-pushing puzzle	tokens	2,500 tokens	7
Search-R1	multi-turn retrieval QA	tokens	3,500 tokens	7
SWE-bench	repository-level bug fixing	tokens	6,772–21,711 (per run)	5
Warehouse	multi-week management sim.	time, capacity, cost	per task, joint	5

run, rather than on coverage. The gate reads the feasibility score and nothing else; a variant that also aborts when spend so far plus the interval's calibrated lower bound exceeds the budget is evaluated on the token environments (Table 3).

4 Experimental setup

Data and environments. All experiments use logged rollouts from BAGEN [13] at a pinned dataset revision; Table 1 summarises the four environments. Sokoban, Search-R1, and SWE-bench are single-resource token tasks. Sokoban and Search-R1 use uniform context budgets of 2,500 and 3,500 tokens; on SWE-bench the official budget varies by run (6,772, 7,060, 8,265, 10,472, and 21,711 tokens across the five models) and we read each run's value from its own record. The label is the number of tokens still needed to complete the task under BAGEN's history-excluded per-turn token accounting. Warehouse is a multi-week management simulation with three jointly binding budgets (time in weeks, capacity in item-weeks, cost in simulated USD), with per-prefix ground truth for all three from the official records; half of its prefixes (640 of 1,280 per model) belong to runs that can no longer finish, which makes it our censoring test.

Protocol, baselines, and models. Rollouts are split 50/25/25 into train, calibration, and test, always by whole rollout. Every primary experiment is repeated over five random splits (seeds 0–4) and reported as the mean; the 19-run breadth count is one split per run, and we say so wherever it appears. Online methods consume the test stream in order with labels revealed only at rollout boundaries. We evaluate three shift scenarios: cross-environment (calibrate on Sokoban; deploy on Search-R1 and SWE-bench), model swap (calibrate on Claude Opus 4.7; deploy on four other agents' SWE-bench streams), and an external-resource model swap on Warehouse. Seven comparators run on the same splits and seeds as the guard: the agent's own interval; a feature-free constant-quantile interval, which measures what the learned component adds; split conformal on absolute residuals; the agent's own midpoint conformalized; the guard with the agent's estimate added to its features; frozen split-CQR under shift; and hard caps for stopping. Coverage is always reported against the 0.90 nominal level and beside the fraction of unbounded intervals. We evaluate seven agents (Claude Opus 4.7/4.6, Claude Sonnet 4.6, GPT-5.2-instant, GPT-5.2-Codex, Gemini 3.1 Pro, and Qwen3-235B), which crossed with environments gives 19 token-budget model$\times$environment runs (7 Sokoban, 7 Search-R1, 5 SWE-bench) of 52–128 rollouts and 85–3,715 prefix samples each. Appendix C gives the full protocol, baseline definitions, metric formulas, and implementation details.

5 Results

5.1 Within-run interval calibration

Figure 3 is the in-distribution picture on Claude Opus 4.7. The agent's own intervals cover 0.51, 0.23, and 0.35 of prefixes on Sokoban, Search-R1, and SWE-bench against a 0.90 target; offline split-CQR reaches 0.92, 0.90, and 0.96, and online windowed-CQR 0.92, 0.92, and 0.87, with no unbounded intervals anywhere. That split-CQR lands near 0.90 when calibration and test are exchangeable is what the construction guarantees; we report it as a check, not a result, and the findings are everything the guarantee does not cover. The intervals are bounded and sharp: windowed-CQR is 22% narrower than the feature-free constant-quantile baseline on Sokoban and 55% narrower on SWE-bench, though 23% wider on Search-R1, where constant-quantile is the sharpest bounded

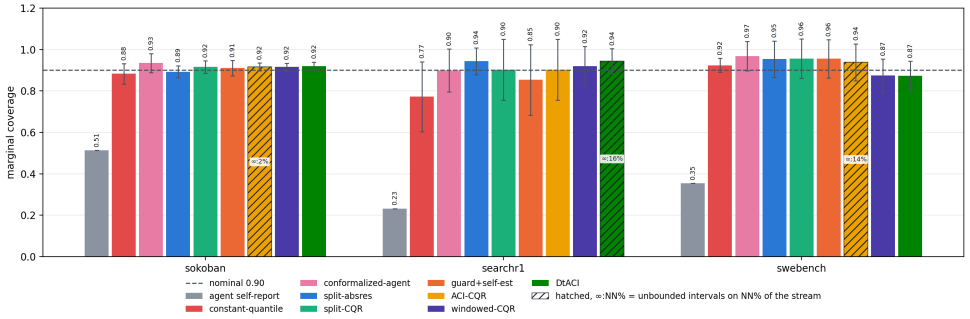


Figure 3: Conformal calibration closes almost all of the gap between agent self-reports and the 0.90 target, and windowed-CQR does so while maintaining bounded intervals throughout. Marginal coverage of the true remaining budget on Sokoban, Search-R1, and SWE-bench (Claude Opus 4.7 rollouts), α=0.10, averaged over five rollout-level split seeds; error bars are the standard deviation across those seeds. Hatched bars emit unbounded intervals on the annotated fraction of the stream, which raises coverage at the cost of informativeness. Agent self-report is a single value read from the logged records rather than a seed mean. Coverage, normalised width, unbounded fraction, and test-set sizes for all methods are in Table 5.

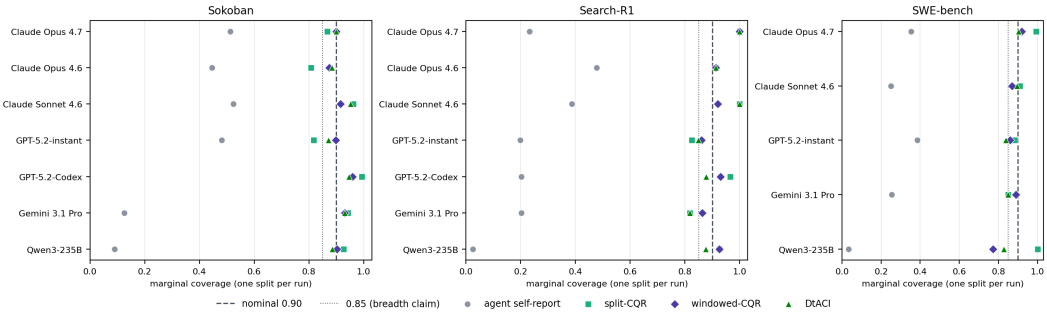


Figure 4: Every token-budget run, one split each (seed 0), no averaging: coverage of the agent's own interval and of split-CQR, windowed-CQR, and DtACI for each of the 19 model–environment runs, against the 0.90 target and the 0.85 line of the breadth claim. Windowed-CQR clears 0.85 on 18 of 19 runs and 0.90 on 11; the agents' own intervals clear neither anywhere. GPT-5.2-Codex on Sokoban carries no agent interval in its records. Table 7 lists the values.

method (Table 5). ACI-CQR and DtACI sometimes post higher coverage, partly by emitting unbounded intervals on up to 16% of the stream. The agent's own estimate adds nothing as a feature: guard+self-est is 0.91 against split-CQR's 0.92 on Sokoban, identical on SWE-bench (0.96), and worse on Search-R1 (0.85 against 0.90). Figure 4 shows every run rather than one agent: across all 19 logged model–environment runs, windowed-CQR reaches at least 0.85 coverage in 18 of 19 and at least 0.90 in 11 of 19, against 14 and 8 for DtACI, on one split per run, and the agents' own intervals clear neither on any run (Table 7). The run of Figure 1 is typical rather than selected: over the 30 test rollouts of that split, split-CQR covers more prefixes than the self-report on 22, ties on 6, and trails on 2 (windowed-CQR: 24, 5, and 1).

5.2 Robustness under deployment shift

This is where the guarantee stops and the experiment starts. Figure 5(a) takes a guard calibrated on Sokoban and points it, unchanged, at Search-R1 and SWE-bench: frozen split-CQR falls to 0.51 and 0.10. Rollout-safe windowed-CQR, recalibrating only from runs that have finished, recovers 0.71 and 0.80 with no unbounded intervals. ACI and DtACI post higher numbers but need unbounded intervals on 60–88% of predictions to get there, which is recovery by surrender rather than by adaptation. Figure 5(b) swaps the agent instead of the task: a guard calibrated on Claude Opus 4.7 is deployed on Sonnet, Gemini, GPT, and Qwen SWE-bench streams. Windowed-CQR reaches 0.66–0.89, below target everywhere, but its seed-to-seed standard deviation is an order of magnitude smaller than frozen calibration's (0.006–0.011 against 0.122–0.224), so its failures are at least predictable. Online adaptation helps under shift and does not solve it: no bounded-interval method

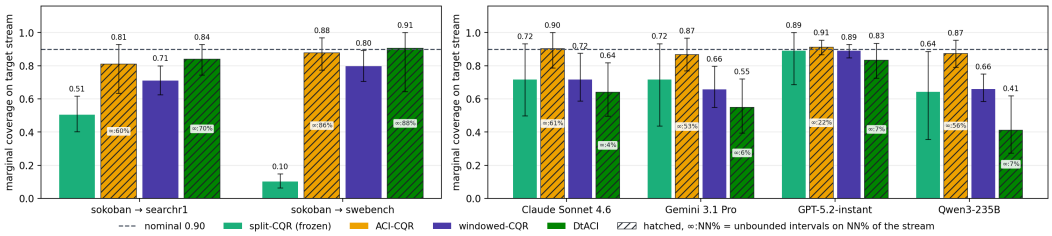


Figure 5: An environment change collapses frozen calibration and online recalibration recovers much of it; a model swap leaves every method below nominal coverage. Marginal coverage on the target stream. (a) A guard calibrated on Sokoban and deployed unchanged on Search-R1 and SWE-bench: frozen split-CQR falls to 0.51 and 0.10, and windowed-CQR recovers to 0.71 and 0.80. (b) A guard calibrated on Claude Opus 4.7 SWE-bench rollouts and deployed on four other agents: here frozen and online calibration are comparable on average (0.74 against 0.73) and neither reaches the target, though windowed-CQR varies far less across seeds. Hatched bars emit unbounded intervals on the annotated fraction of the stream (60–88% for the ACI-based methods in (a), and 22–61% (ACI-CQR) and 4–7% (DtACI) in (b)).

recovers the nominal level under a model swap, and any coverage number has to be read next to how many of its intervals were infinite.

5.3 Conditional and multi-resource calibration

A marginal number can hide a hole in the slice that matters. On Sokoban, split-CQR reaches 0.92 marginal coverage but only 0.86 at the first turn, exactly where an abort decision has the most leverage; Mondrian calibration by turn depth raises first-turn coverage to 0.93, is neutral to slightly negative on Search-R1, and neutral to slightly positive on SWE-bench, where the first-turn gap is already small (Table 8). What it buys is conditional validity, not a uniform improvement.

Table 2: Independent margins under-cover the joint event, Bonferroni fixes it in distribution, and nothing with a guarantee survives a model swap. Warehouse coverage against a nominal 0.90, mean over five rollout-level split seeds; above 0.90, coverage is bought with wider margins. Per-resource columns and the in-distribution joint column are Claude Opus 4.7; the final column is the joint event averaged over four swap targets. Bold marks the highest joint coverage in each column. "–" marks a scheme not run in that setting.

	per-resource			joint	
Calibration scheme	time	capacity	cost	in-dist.	model swap
agent self-report	0.87	0.67	0.71	0.42	0.34
independent margins (α)	0.99	0.89	0.92	0.80	–
Bonferroni ($\alpha/3$)	1.00	0.98	0.97	**0.95**	0.71
empirical copula	0.99	0.94	0.96	0.89	0.62
adaptive copula (online)	–	–	–	–	**0.87**

Table 2 is the multi-resource case. In Warehouse the agent's self-reports cover the joint event across time, capacity, and cost 0.42 of the time. Calibrating each resource independently at α lifts that only to 0.80: independent margins do not compose into joint coverage. Bonferroni at $\alpha/3$ restores 0.95, and the empirical copula reaches 0.89 with narrower margins but no guarantee. Under an external-resource model swap, frozen Bonferroni falls to 0.71, below its nominal level and below the 0.87 of the un-guaranteed adaptive copula, which recalibrates from finished rollouts. We report both and rest the claim on the one with the guarantee.

5.4 Economic impact

The gate is judged on money, not coverage. Figure 6 replays Warehouse with the classifier-only gate at δ=0.10: it avoids 86.3% of doomed-run spend with 0% observed false aborts. Hard spending

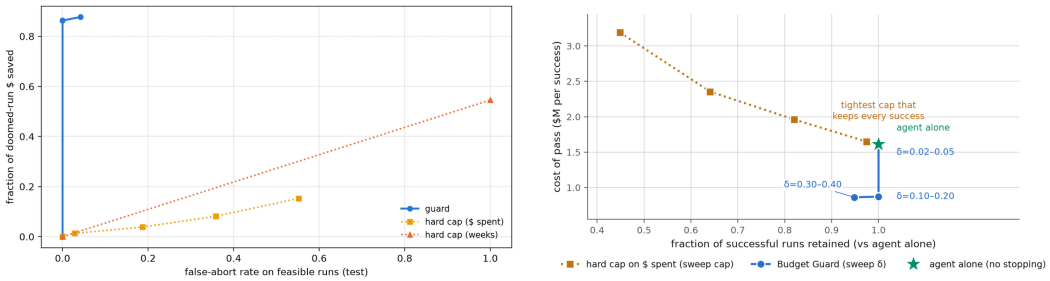


Figure 6: Warehouse economic impact from the feasibility gate alone, without remaining-budget intervals. At δ=0.10 the classifier-only rule avoids 86.3% of doomed-run spend with no observed false aborts, which no hard cap achieves at any setting. Simulated USD, mean over five seeds. (a) Every hard cap that saves more than 0.2% of doomed spend also stops runs that would have succeeded; the gate occupies the favourable region of the savings–false-abort trade-off. (b) Cost-of-Pass against the fraction of successful runs retained: the gate roughly halves cost per pass while retaining every observed success, whereas no cap improves on the unstopped baseline. Both panels are counterfactual replays of logged rollouts, not live deployments.

caps cannot: the tightest dollar cap that preserves every successful run saves 0.2% of doomed spend, the dollar caps that save 4–15% abort 19–55% of the runs that would have succeeded, and the only week cap that saves anything (54.5%) aborts every feasible run. The gate cuts Cost-of-Pass from \$1.61M to \$0.87M, a reduction of 46%, while retaining every observed success; no cap improves on the unstopped baseline by more than a rounding error, and the tightest raises cost per pass to \$3.19M by discarding passes.

Table 3: Abort performance at δ=0.10 (mean over five seeds). Warehouse uses the classifier-only gate (native USD). Token abort is evaluated on Sokoban and SWE-bench only; both also evaluate a classifier+interval variant. Combining intervals reduces doomed savings at this operating point.

Environment	Rule	False abort	Doomed spend saved	Uses intervals?
Warehouse	feasibility only	0.0%	86.3%	No
Sokoban	feasibility only	5.8%	7.1%	No
Sokoban	feasibility+interval	5.8%	1.9%	Yes
SWE-bench	feasibility only	0.0%	3.1%	No
SWE-bench	feasibility+interval	0.0%	0.0%	Yes

Table 3 puts the same gate on the token tasks, and the picture changes. At the same δ=0.10, Sokoban and SWE-bench save only 7.1% and 3.1% of doomed token spend, with false-abort rates of 5.8% and 0%. Adding the interval trigger to the rule makes it more conservative at fixed δ (Sokoban savings fall to 1.9%, SWE-bench to 0%) and recovers none of the Warehouse-scale savings. This is the result behind keeping the two components apart. The intervals measure how much a run still needs, and §5.1 shows they measure it reliably; they are not a stopping signal, and used as one they only make the gate timid. What they are for is planning and monitoring: a deployer reads spend so far plus the upper bound against the budget to see whether a run still fits, as in Figure 1, or sizes the quota for a batch of runs before launching it. The gate is for stopping. The 86.3% is therefore a Warehouse result for the gate alone, not a general Budget Guard savings rate. Two further cautions: the dollars are simulated Warehouse dollars, not comparable to BAGEN's 28–64% token savings; and an observed 0% false-abort rate on a small test set is consistent with the δ=0.10 bound, not proof that the true rate is zero.

5.5 Censored calibration

Roughly half of Warehouse prefixes are censored: the run can no longer finish, so what it would have needed is never observed. Treating those runs as missing data is the wrong call. Dropping them yields only 0.50 coverage on the censored prefixes where cost is the binding resource, which are exactly the expensive failures. Treating censored scores as $+\infty$ is valid but vacuous at this

censoring rate, and the informative treatments trade validity for sharpness in ways Table 10 makes explicit. The other lever is to abstain: feasibility-gated selective intervals hold coverage at 0.92–0.95 for capacity and cost, and 1.00 for time, on the prefixes they keep, at the cost of keeping between 51% and 89% of them (Table 9). That gate is swept over its own prefix-level tolerance δ_{gate}, a different quantity from the rollout-level δ of §5.4. One caveat governs these numbers: the recovered labels used to score censored prefixes are exact only for time, and carry a week-proportional interpolation with 12–27% median relative error for capacity and cost, so those two are coverage against an approximate label rather than ground truth.

6 Discussion and limitations

What the paper does not show matters as much as what it does. It does not show a general savings rate: the 86.3% is Warehouse, gate-only, and falls to 7.1% and 3.1% on token tasks at the same δ (Table 3), and the gate pays only where doomed runs are expensive and identifiable early. The joint results rest on one multi-resource environment, which cannot support a claim that they generalise. The in-distribution coverage numbers are a check on the construction, not evidence for the method; the evidence is in the shift, sharpness, joint, censoring, and economic results.

The breadth count is a single split: on seed 0, windowed-CQR reaches at least 0.85 coverage in 18 of 19 runs. Search-R1 leaves about 24 test prefixes per seed, which is why its standard deviations reach 0.10–0.17. Coverage alone does not distinguish a useful interval from a trivially wide one, so width and unbounded fraction are reported throughout; Figure 1 also shows the price of calibration, intervals about one and a half times as wide as the agent's, which costs nothing when the question is whether the run still fits and matters when a tight point estimate is needed. Censored capacity and cost results are measured against an interpolated label. We have not compared against an agent trained to be budget-aware, such as BAGEN's fine-tuned estimator, because its checkpoint is not released. Finally, savings and Cost-of-Pass are counterfactual replays: they assume adding the guard would not change agent behaviour and that the logged sample is representative. A live deployment, in which the agent sees the guard's interval and the gate actually stops runs, is the experiment that would settle both assumptions, and we have not run it.

Broader impact. The intended effect is to reduce wasted compute, energy, and money on agent runs that cannot succeed; the Warehouse gate at $\delta{=}0.10$ illustrates that potential in one environment (86.3% of doomed-run spend avoided in replay, without using remaining-budget intervals). The corresponding risk is that a stopping threshold set too aggressively terminates runs that would have finished; δ calibration exists to bound that risk at a user-chosen level, and the counterfactual caveat above means a deployer should not treat the reported savings as a guaranteed live return when choosing δ, nor assume token environments will see Warehouse-scale savings.

7 Conclusion

Budget awareness can be supplied from outside the agent. A training-free layer over the usage counts an API already reports turns them into remaining-budget intervals with 0.90–0.96 coverage offline and 0.87–0.92 online on Claude Opus 4.7 against a 0.90 target, and at least 0.85 in 18 of 19 model$\times$environment runs on one split each, where the agent's own intervals manage 0.23–0.51. That guarantee assumes exchangeability, and the experiments map what happens without it: online recalibration recovers much of an environment shift, no bounded method reaches nominal under a model swap, and joint calibration works in distribution and not under swap. A separately calibrated feasibility gate avoids 86.3% of doomed-run spend in Warehouse replay at $\delta{=}0.10$, and far less on token tasks. Budget awareness is a measurement and a decision, and it works when the two are calibrated, evaluated, and claimed separately.

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A Background and guarantees

This appendix defines the concepts needed to interpret the results.

- **Coverage.** The fraction of cases in which the true remaining budget falls inside the predicted interval. Coverage of 0.90 means the true value is contained in about 90% of cases.
- **Calibration set.** A held-out set of completed trajectories used to set interval widths and stopping thresholds.
- **Rollout (run).** One complete agent attempt at a task.
- **Turn (k).** One agent step within a rollout.
- **Prefix.** The partial trajectory containing all turns completed through turn k.
- $\hat{q}$**.** The conformal threshold: a learned error margin added to the model's predictions to expand them into an interval. Throughout, $\hat{q}$ is the conformal threshold and $\hat{f}_{\mathrm{lo}}$, $\hat{f}_{\mathrm{hi}}$ are the quantile regressor's own predictions.
- α**.** The allowed miscoverage rate; the target coverage is $1-\alpha$. We use $\alpha{=}0.10$ throughout. A smaller α raises the target and generally widens intervals.
- **Width.** The size of the predicted range. Narrower is more informative at equal coverage: $[100, 115]$ is more useful than $[100, 300]$. An interval extending to infinity attains coverage while saying nothing.
- δ**.** The false-abort tolerance for the stopping rule. A larger δ permits greater savings on doomed runs while accepting more risk of aborting successful ones.
- τ**.** The abort threshold. The guard aborts at the first turn whose feasibility score falls below τ. It is not chosen by hand: it is read off the calibration rollouts as the value holding the false-abort rate at or below δ, so a smaller δ produces a more cautious τ.
- **Offline.** Calibrates once on completed runs and stays fixed during deployment.
- **Online.** Updates calibration after each completed run, using its newly observed remaining-budget labels.

Table 4 states, component by component, what is guaranteed and under what assumption. Four rows carry no finite-sample guarantee at all (windowed-CQR, the empirical copula, the censoring-aware bounds other than $+\infty$, and the marginal rather than selective reading of the gated intervals), and every claim resting on those is reported in this paper as an empirical observation.

B Per-run breadth results

Table 7 lists every model×environment run behind the breadth claims in §5.1; the "18 of 19" and "11 of 19" counts can be read straight off the windowed-CQR column. Table 8 gives conditional coverage by turn depth, Table 9 the feasibility-gate sweep, and Table 10 the censored upper-bound calibrations. Every number is read from the generated result tables, which one command regenerates and two checkers verify. Table 6 gives the numbers behind Figure 1.

Table 4: What each component guarantees, and what it does not. Statements we do not have are listed as such, so a reader can tell a theorem from an observation.

Method	Role in Budget Guard	Guarantee and scope
split-CQR	CQR intervals conformalized on a by-rollout calibration split; our offline default [10].	Marginal coverage $\geq 1-\alpha$ over a random (rollout, prefix) pair, under exchangeable rollouts. Pooling a rollout's prefixes is an empirical evaluation, not the guaranteed object [15].
Mondrian-CQR	split-CQR calibrated separately per turn-depth bucket [20].	Coverage $\geq 1-\alpha$ within each bucket, under exchangeability within the bucket; thin buckets fall back to the global quantile.
ACI / DtACI	Adaptive coverage levels recalibrated from the stream's finished rollouts [11, 12].	Long-run average coverage approaches $1-\alpha$. May emit unbounded intervals; we clamp the working level at 0.999 rather than emit empty sets, a conservative deviation.
windowed-CQR	Conformal quantile over a sliding window of the 100 most recent scores; an SPCI-inspired simplification [18].	No finite-sample guarantee; coverage gap bounded by window-to-stream drift [19]. Evaluated empirically.
Bonferroni joint	Calibrates each resource at $\alpha/3$ so all hold at once; our primary multi-resource calibration scheme.	Joint coverage $\geq 1-\alpha$ by the union bound; no data-driven selection; conservative, so realised coverage exceeds the target in distribution.
empirical copula	Shared copula rank across resources [21]; an adaptive variant updates the rank per finished rollout.	Empirical only: the rank is selected on the same scores it thresholds. Reported alongside Bonferroni, not as the guaranteed claim.
censoring-aware upper bounds	One-sided bounds for doomed runs [22–24].	One-sided upper coverage on the censored subset. $+\infty$ always valid but vacuous once the censored fraction exceeds α; the others empirical or assumption-dependent.
feasibility-gated intervals	A classifier on the same prefix features issues intervals only on predicted-feasible prefixes.	Coverage $\geq 1-\alpha$ conditional on passing the gate: selective, not marginal. Marginal coverage reported empirically.
stopping rule (τ)	Abort at the first turn whose feasibility score drops below τ [26]; scored by Cost-of-Pass [25].	False-abort probability $\leq \delta$ per run, from rollout-level calibration on feasible runs.

C Protocol, baselines, metrics, and implementation

This appendix records what §4 summarises: how the data is split, what each baseline is, how each reported number is computed, and the settings under which all of it runs. Every value below is fixed in the released code, and the parenthetical names are the modules that implement it.

C.1 Splits and seeds

Every split is by rollout, never by prefix (`split.group_split`). For a given seed the unique rollout indices are permuted with `numpy.random.default_rng(seed)` and allocated 50/25/25 to train, calibration and test; every prefix follows its rollout. Prefixes of one rollout share that rollout's final total, so splitting them apart would place each test rollout's answer into calibration. A split that is requested but comes out empty raises an error rather than degrading quietly: with an empty calibration set the conformal quantile of zero scores is $+\infty$, every interval becomes $[0, \infty)$, and coverage reads a meaningless 1.00 with no other symptom. Reported numbers are means over five seeds, 0 through 4.

C.2 Rollout-safe online evaluation

Online methods consume the test rows in their natural trajectory order, rollout by rollout and turn by turn. A prefix's label is the resource the run still needs to finish, which cannot be observed until the rollout ends, so a rollout's labels enter the update state only at its boundary and the state is frozen within a rollout (`conformal.aci_cqr`, `windowed_cqr`, `dtaci`, each taking a `groups` argument). The code rejects test streams whose rollout ids are not contiguous blocks: an id that reappears after a boundary would silently absorb a still-running rollout's labels and inflate coverage with no visible symptom. Revealing labels per prefix instead would use information a deployed guard does not

Table 5: Full within-run results: every conformal method lands within 0.05 of the 0.90 target except the feature-free constant-quantile baseline on Search-R1, where the agent's own intervals miss by 0.39–0.67; windowed-CQR gives the narrowest bounded intervals on Sokoban and the second-narrowest on SWE-bench, while constant-quantile is sharpest on Search-R1. Claude Opus 4.7, nominal 0.90, mean$\pm$ standard deviation over five rollout-level split seeds. Normalised width is interval width divided by the central 80% range of the labels (lower is sharper). "Unb." is the percentage of predictions with an infinite upper endpoint; width is shown as "–" where at least one seed produced a majority of unbounded intervals. n_{test} is test prefixes per seed.

Env	Method	Coverage	Norm. width	Unb. (%)	n_{test}
Sokoban	agent self-report	0.51	–	–	–
	constant-quantile	0.88 ± 0.05	1.10	0	87
	split-absres	0.89 ± 0.03	1.12	0	87
	split-CQR	0.92 ± 0.03	0.88	0	87
	conformalized-agent	0.93 ± 0.05	1.44	0	87
	guard+self-est	0.91 ± 0.04	0.91	0	87
	ACI-CQR	0.92 ± 0.02	0.88	2	87
	windowed-CQR	0.92 ± 0.02	0.86	0	87
	DtACI	0.92 ± 0.02	0.88	0	87
Search-R1	agent self-report	0.23	–	–	–
	constant-quantile	0.77 ± 0.17	0.90	0	24
	split-absres	0.94 ± 0.06	1.60	0	24
	split-CQR	0.90 ± 0.15	1.14	0	24
	conformalized-agent	0.90 ± 0.10	1.96	0	24
	guard+self-est	0.85 ± 0.17	1.07	0	24
	ACI-CQR	0.90 ± 0.15	1.14	0	24
	windowed-CQR	0.92 ± 0.10	1.10	0	24
	DtACI	0.94 ± 0.06	–	16	24
SWE-bench	agent self-report	0.35	–	–	–
	constant-quantile	0.92 ± 0.03	2.87	0	161
	split-absres	0.95 ± 0.09	2.56	0	161
	split-CQR	0.96 ± 0.10	2.11	0	161
	conformalized-agent	0.97 ± 0.07	2.23	0	161
	guard+self-est	0.96 ± 0.09	1.97	0	161
	ACI-CQR	0.94 ± 0.09	–	14	161
	windowed-CQR	0.87 ± 0.08	1.29	0	161
	DtACI	0.87 ± 0.07	1.22	0	161

Table 6: The first held-out Sokoban rollout in the seed-0 test stream, turn by turn (Claude Opus 4.7, 2,500-token budget); the run plotted in Figure 1. $\surd$: the interval contains the true remaining spend; $\times$: it misses. "Spent + upper" is spend so far plus the guard's upper bound; "Gate" is the feasibility score and decision at δ=0.10 (abort below 0.02).

Turn	Spent	True rem.	Agent self-report	Guard interval (split-CQR)	Spent + upper	Gate
1	557	1,090	[1,000, 1,800] $\surd$	[460, 1,582] $\surd$	2,139 $\leq$ 2,500	0.99 continue
2	1,076	571	[600, 1,200] $\times$	[210, 1,102] $\surd$	2,178 $\leq$ 2,500	0.95 continue
3	1,380	267	[400, 900] $\times$	[162, 902] $\surd$	2,282 $\leq$ 2,500	0.99 continue

have, so we report only the rollout-safe protocol. An earlier revision of this evaluation revealed labels per prefix; correcting it lowered the headline cross-environment recovery, windowed-CQR on SWE-bench (§5.2), from 0.90 to 0.80.

C.3 Base predictor and features

The point and quantile predictors are LightGBM models [27] with `n_estimators`=200, `learning_rate`=0.05, `num_leaves`=15 and `min_child_samples`=10. CQR fits two quantile regressors, at $\alpha/2$ and $1-\alpha/2$; the absolute-residual baseline and the interval-driven stopping variant use a squared-error regressor; the feasibility gate is an LGBM classifier with the same settings.

Table 7: Breadth across all 19 token-budget runs: windowed-CQR clears 0.85 on 18 of them and 0.90 on 11, while the agents' own intervals clear neither anywhere. Marginal coverage against nominal 0.90 on a single rollout-level split (seed 0). Four decimals are shown on the three guard columns so the two counts above are directly checkable; at three decimals SWE-bench/Gemini DtACI rounds to 0.850 and appears to clear 0.85 when it does not. "–" marks a run whose records carry no agent interval. †Warehouse is listed for dataset composition only; its 0.424 is joint coverage across three resources (Table 2), not a token-budget marginal.

Environment	Model	Rollouts	Prefixes	Agent	split-CQR	windowed-CQR	DtACI
Sokoban	Claude Opus 4.6	116	410	0.446	0.8077	0.8750	0.8846
	Claude Opus 4.7	123	355	0.513	0.8667	0.9000	0.9000
	Claude Sonnet 4.6	126	400	0.524	0.9623	0.9151	0.9528
	Gemini 3.1 Pro	126	389	0.125	0.9425	0.9310	0.9310
	GPT-5.2-instant	128	661	0.481	0.8176	0.8986	0.8716
	GPT-5.2-Codex	128	592	–	0.9933	0.9600	0.9467
	Qwen3-235B	126	1,038	0.090	0.9268	0.9024	0.8862
Search-R1	Claude Opus 4.7	66	87	0.233	1.0000	1.0000	1.0000
	Claude Sonnet 4.6	64	102	0.386	1.0000	0.9200	1.0000
	Claude Opus 4.6	52	90	0.478	0.9130	0.9130	0.9130
	GPT-5.2-instant	125	324	0.198	0.8256	0.8605	0.8488
	GPT-5.2-Codex	126	262	0.202	0.9649	0.9298	0.8772
	Gemini 3.1 Pro	83	85	0.202	0.8182	0.8636	0.8182
	Qwen3-235B	124	1,079	0.025	0.9259	0.9259	0.8765
SWE-bench	Claude Opus 4.7	64	712	0.354	0.9926	0.9185	0.9037
	Claude Sonnet 4.6	64	1,258	0.251	0.9101	0.8705	0.8957
	Gemini 3.1 Pro	64	2,327	0.254	0.8497	0.8899	0.8497
	GPT-5.2-instant	64	1,313	0.385	0.8840	0.8589	0.8370
	Qwen3-235B	60	3,715	0.034	1.0000	0.7727	0.8271
Warehouse	Opus 4.7 + 4 swap targets	64 ea.	1,280 ea.	0.424†	–	–	–

Table 8: Marginal coverage hides a first-turn deficit on Sokoban, and Mondrian calibration is what removes it. Coverage by turn-depth bucket, nominal 0.90, mean over five seeds. Search-R1 rollouts are short enough that no $k{\geq}3$ bucket exists.

Environment	Method	Marginal	$k{=}1$	$k{=}2$	$k{\geq}3$
Sokoban	split-CQR	0.92	0.86	0.93	0.96
	Mondrian-CQR	0.94	0.93	0.96	0.94
Search-R1	split-CQR	0.90	0.91	0.89	–
	Mondrian-CQR	0.89	0.89	0.89	–
SWE-bench	split-CQR	0.96	0.99	0.99	0.95
	Mondrian-CQR	0.97	0.99	0.99	0.96

No hyperparameter search was run: these are the defaults fixed at the start of the project and used unchanged for every experiment, baseline included.

The feature matrix (`features.build_xy`) is: the turn index k; cumulative spend; burn rate (spend/k); the last, mean, standard deviation, minimum and maximum per-turn spend; the mean of the last three turns; and the difference between the last turn and the mean. When the agent emits its own interval, four columns are added (the two endpoints, the midpoint and the width), and a missing estimate is left as `NaN`, which LightGBM handles natively. Quantities known only at rollout end (final totals, success, feasibility) are listed in a leak set and asserted absent from the matrix at build time, so the exclusion is enforced rather than intended.

C.4 Conformal methods and their settings

Every method uses the finite-sample corrected quantile: with n calibration scores the level is $\lceil (n+1)(1{-}\alpha) \rceil / n$, read with higher interpolation, returning $+\infty$ when that level exceeds 1. All intervals are clipped below at zero, since a remaining budget cannot be negative. $\alpha{=}0.10$ throughout.

Table 9: The feasibility gate trades retained volume for coverage on what it keeps: abstaining on predicted-doomed prefixes holds 0.92–0.95 coverage on capacity, where marginal calibration reaches only 0.89. Warehouse, mean over five seeds. "Kept" is the fraction of prefixes the gate admits, "caught" the fraction of doomed prefixes it correctly abstains on, "false abort" the fraction of feasible prefixes it wrongly rejects, and "coverage" the interval coverage among admitted feasible prefixes. Time binds in only 30% of censored runs and the gate catches almost all of them, so its row is flat. δ here is the gate's prefix-level tolerance, not the rollout-level false-abort tolerance of §5.4.

Resource	δ	Kept (↑)	Caught (↑)	False abort (↓)	Coverage (↑)
time (weeks)	0.02	0.88	0.93	0.00	1.00
	0.05	0.88	0.93	0.00	1.00
	0.10	0.88	0.93	0.00	1.00
	0.20	0.88	0.93	0.00	1.00
	0.30	0.88	0.93	0.00	1.00
capacity (item-weeks)	0.02	0.89	0.31	0.04	0.95
	0.05	0.78	0.64	0.05	0.95
	0.10	0.70	0.74	0.07	0.95
	0.20	0.57	0.82	0.19	0.95
	0.30	0.51	0.85	0.23	0.94
cost (USD)	0.02	0.86	0.60	0.05	0.92
	0.05	0.81	0.69	0.10	0.92
	0.10	0.80	0.72	0.10	0.92
	0.20	0.74	0.75	0.21	0.93
	0.30	0.74	0.76	0.21	0.93

split-CQR uses the score $\max(\hat{f}_{\text{lo}} - y,\ y - \hat{f}_{\text{hi}})$ and one global $\hat{q}$; it is the offline default. Mondrian-CQR applies the same score with one $\hat{q}$ per turn-depth bucket, split at $k{=}1$, $k{=}2$ and $k{\geq}3$; a bucket holding fewer than 10 calibration scores falls back to the global $\hat{q}$.

ACI updates the working level by $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$. γ is selected on calibration data only: the calibration rollouts are split in half, ACI is streamed over the second half under the same rollout-safe revelation used at test time, and the $\gamma \in \{0.005, 0.01, 0.02, 0.05\}$ whose coverage lands closest to $1{-}\alpha$ is kept. One deliberate deviation from canonical ACI: where the canonical algorithm emits the empty set once $\alpha_t \geq 1$, we clamp the quantile lookup at 0.999 and emit a finite interval. The deviation is conservative in both coverage and the error feedback, and we state it rather than diverge silently.

windowed-CQR keeps a rolling window of the 100 most recent conformity scores, warm-started from calibration, and reads the interval half-width off that window's empirical quantile, so the score distribution itself tracks drift rather than reweighting a frozen calibration set. SPCI [18] is the inspiration, not the implemented method.

DtACI aggregates ACI experts at $\gamma \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$ with learning rate $\eta = 2.0$ and uniform mixing $\sigma = 0.01$. After each revealed label, experts are scored by pinball loss against β_t, the largest miscoverage level whose quantile would still have covered the observed score, and the working level is the weight-averaged α_t.

C.5 Baselines

Seven comparators run on the same splits and seeds as the guard.

1. **Agent self-report.** BAGEN's own per-sample indicators of whether the agent's interval contained the truth, read from the official estimation records. No calibration is applied.
2. **Constant-quantile.** Feature-free: the empirical $[\alpha/2, 1{-}\alpha/2]$ interval of the calibration labels, identical at every test prefix.
3. **Split-absres.** A LightGBM point prediction plus or minus the calibrated absolute-residual quantile.
4. **Conformalized-agent.** The agent's own interval midpoint used as the point predictor and conformalized on $|y - \text{mid}|$.

Table 10: Dropping censored runs under-covers exactly the expensive failures, and every informative repair trades validity for sharpness. One-sided upper-bound coverage on Warehouse at nominal 0.90, split by whether the resource was the binding constraint. $+\infty$ is always valid and always vacuous; "drop" is the naive comparison. Kaplan–Meier is a diagnostic, not a calibration option. Coverage for capacity and cost is measured against an interpolated label (12–27% median relative error), not ground truth.

Resource	Treatment	Feasible (↑0.90)	Cens. binding	Cens. non-binding	Binding frac.
time (weeks)	drop (naive)	1.00	1.00	1.00	0.30
	$+\infty$	1.00	1.00	1.00	0.30
	substituted bound	1.00	1.00	1.00	0.30
	IPCW	0.96	0.96	0.96	0.30
	per-resource bound	0.90	0.90	0.90	0.30
	Kaplan–Meier (diagnostic)	1.00	1.00	1.00	0.30
capacity (item-weeks)	drop (naive)	0.94	0.77	0.98	0.34
	$+\infty$	1.00	1.00	1.00	0.34
	substituted bound	0.99	0.98	1.00	0.34
	IPCW	0.93	0.75	0.98	0.34
	per-resource bound	0.91	0.74	0.98	0.34
	Kaplan–Meier (diagnostic)	0.94	0.78	0.99	0.34
cost (USD)	drop (naive)	0.88	0.50	0.97	0.42
	$+\infty$	1.00	1.00	1.00	0.42
	substituted bound	1.00	0.99	1.00	0.42
	IPCW	0.88	0.50	0.96	0.42
	per-resource bound	0.86	0.46	0.94	0.42
	Kaplan–Meier (diagnostic)	0.92	0.58	0.98	0.42

5. **Guard+self-est.** split-CQR on the guard features augmented with the agent's estimate columns.
6. **Frozen split-CQR.** split-CQR calibrated on the source environment or model and applied unchanged to the target stream.
7. **Hard caps.** Abort once spend so far exceeds a fixed cap. Dollar caps are swept at the 20th, 35th, 50th, 65th and 80th percentiles of realised run totals, and week caps at $0.5\times$ the 25th, $0.75\times$ the 50th and $1.0\times$ the 75th percentile of the task's time budget.

Every baseline sees the same training split, the same calibration split, the same test stream and the same seeds as the guard. BAGEN's SFT+RL estimator is not included as a baseline: it is a fine-tuned agent rather than an external wrapper, and re-running it would require training the underlying LLM, which is exactly what this work avoids.

C.6 Metrics

Coverage is the fraction of test prefixes whose interval contains the true remaining requirement. Normalised width is the interval width divided by the central 80% range of the labels, $q_{0.9}-q_{0.1}$, which makes environments whose token scales differ by orders of magnitude comparable; the median width is used wherever a method can emit unbounded intervals, and the across-seed summary is reported as "–" where at least one seed's median is itself infinite. Unbounded fraction is the share of predictions with an infinite upper endpoint, and it appears beside every coverage number.

Confidence intervals come from a cluster bootstrap that resamples whole rollouts rather than prefixes, with $B = 500$ resamples; per-sample binomial intervals are too tight here, because the prefixes of one rollout are correlated. Across seeds, the per-seed bootstrap draws are pooled and the 5th and 95th percentiles of the pooled distribution are reported as a 90% interval; these are recorded in the released result tables rather than plotted. The $\pm$ shown in Table 5 and the error bars in Figure 3 are both the standard deviation across the five seeds.

For the stopping rule, the false-abort rate is the fraction of feasible test rollouts that were stopped, and the doomed-savings fraction is the spend avoided on rollouts that could not finish divided by the spend those rollouts would otherwise have incurred. Cost-of-Pass is total spend under a policy

divided by the number of runs that still pass [25]. All economic results in this paper are Warehouse, which is priced natively in simulated dollars; token-environment economics appear only in Table 3.

C.7 Multi-resource and censored calibration

Warehouse binds time in weeks, capacity in item-weeks and cost in simulated dollars at once. Independent calibration sets each margin at α; Bonferroni sets each at $\alpha/3$, which restores a joint guarantee of $1-\alpha$ by the union bound and is the primary multi-resource calibration scheme. The empirical-copula variant shares one rank across resources and is tighter, but that rank is selected on the same calibration scores it then thresholds, so it carries no guarantee and is reported as an observation only. Under a model swap, an adaptive variant recomputes both the margins and the shared rank from a rolling window of recent three-resource error vectors, updated only once a rollout has finished.

For censored runs the one-sided upper bound is calibrated five ways (`censored.upper_qhat`): drop; infinity; bound; ipcw; and resource_bound. A Kaplan–Meier estimator [24] is a diagnostic rather than a sixth option: it returns $+\infty$ exactly when the upper tail is entirely censored. Table 10 reports the five calibrations together with the Kaplan–Meier diagnostic. Censored-stratum coverage for capacity and cost is coverage against an approximate interpolated label (12–27% median relative error), not against ground truth.

The feasibility gate is an LGBM classifier on the same prefix features. It issues intervals only on predicted-feasible prefixes, so the resulting guarantee is selective. The gate is swept at its own prefix-level tolerance $\delta_{\text{gate}} \in \{0.02, 0.05, 0.10, 0.20, 0.30\}$, which is a different quantity from the rollout-level false-abort tolerance δ of §5.4; Table 9 reports the full sweep.

C.8 Stopping rule and economics

The reported rule aborts at the first turn whose feasibility score falls below τ. Because the rule aborts a rollout exactly when that rollout's minimum prefix score crosses τ, τ is calibrated at the rollout level: it is set to the score at rank $\lfloor (n+1)\delta \rfloor$ among the sorted minimum scores of the n feasible calibration rollouts, which bounds by δ the probability of aborting a run that would have finished. δ is swept over $\{0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}$ for the frontier; $\delta{=}0.10$ is the operating point quoted in §5.4 and the abstract.

An interval-driven variant, which aborts when spend so far plus the calibrated lower bound exceeds the budget, is implemented in the released code and calibrated the same way at rollout level. It is not the rule behind the reported Warehouse economics, and we do not claim the two are equivalent. The economic numbers are counterfactual replays, not live measurements: each completed run is truncated where the guard would have aborted it and the avoided spend is summed.

C.9 Data, environment, and reproduction

All experiments use logged rollouts from BAGEN [13], fetched from its public Hugging Face release at the pinned revision `58189576e54b675fdd0e1d6c1c9f189c2992732f`. Its dataset card states no licence, so we use the data as publicly released for research and credit the originators rather than asserting terms the release does not state. The extractor reproduces BAGEN's turn-excluding-history token accounting exactly, with 0 of 355 mismatches against the official per-sample ground truth on the reference run.

The whole pipeline runs inside a container whose base image is pinned by digest (`python:3.12-slim-bookworm`), with numpy 2.5.1, pandas 3.0.3, LightGBM 4.6.0, scikit-learn 1.9.0, scipy 1.18.0 and matplotlib 3.11.0 resolved from a hash-locked requirements file. LightGBM is MIT-licensed; numpy, pandas, scipy and scikit-learn are BSD-3-Clause; matplotlib uses a PSF-style licence. The guard is CPU-only: no GPU, no model weights, and no access to logits or environment internals. No LLM is trained, fine-tuned or queried by us at any stage. A single command regenerates every figure and table in the repository and then machine-verifies the paper against what it just produced, exiting non-zero if any reported claim no longer matches.

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